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A Proofs for connectivity of dropout-stable optima

Proof of Lemma 1. Without loss of generality, suppose for each θ_i that the subset of $\lfloor h_i/2 \rfloor$ non-zero hidden units in each layer are all indexed between 1 and $\lfloor h_i/2 \rfloor$. For $1 < i < d$, we can partition A_i into quadrants such that $A_i = \left[\begin{array}{c|c} L_i & C_i \\ \hline D_i & R_i \end{array}\right]$. (Here, $L_i \in \mathbb{R}^{\lfloor h_i/2 \rfloor \times \lfloor h_i/2 \rfloor}$. If h_i is odd, when we write L_i in the other quadrants we implicitly pad it with zeros in a consistent manner.) Similarly, we can partition A_1 such that $A_1 = \left[\begin{array}{c} L_1 \\ \hline B_1 \end{array}\right]$ and A_d such that $A_d = \left[\begin{array}{c|c} L_d & R_d \end{array}\right]$. We will sometimes use the notation A_i to refer to the value of A_i at a given point on our path, while A_i^θ will always refer to the value of A_i at θ. We now proceed to prove via induction the existence of a path from θ to θ_i for all i whose loss is bounded by $L(f_\theta) + \epsilon$, from which the main result immediately follows.

Base case: from θ to θ_{d-1} As a base case of the induction, we need to construct a path from θ to θ_{d-1}, such that the loss is bounded by $L(f_\theta) + \epsilon$. First, note that setting a particular subset of columns (e.g. the right half of columns) in A_i to zero is equivalent to setting the corresponding rows (e.g. the bottom half of rows) of A_{i-1} to zero. So from the fact that $L(f_{\theta_{d-1}}) \leq L(f_\theta) + \epsilon$ it follows that we can equivalently replace A_d^θ with $\left[\begin{array}{c|c} rL_d^\theta & 0 \end{array}\right]$ without increasing our loss by more than ϵ.

In fact, because our loss function is convex over A_d we can actually interpolate A_d between A_d^θ and $\left[\begin{array}{c|c} rL_d^\theta & 0 \end{array}\right]$ while keeping our loss below $L(f_\theta) + \epsilon$ at every point along this subpath.

Then, because $R_d = 0$ we can modify both D_{d-1} and R_{d-1} any way we'd like without affecting the output of our network. In particular, we can interpolate A_{d-1} between A_{d-1}^θ and $\left[\begin{array}{c|c} L_{d-1}^\theta & C_{d-1}^\theta \\ \hline 0 & 0 \end{array}\right]$ while keeping our loss constant long this subpath, thus arriving at θ_{d-1}.

From θ_k to θ_{k-1}

Suppose we have found a path from θ to θ_k such that (1) $A_d^{\theta_k} = \left[\begin{array}{c|c} rL_d^\theta & 0 \end{array}\right]$, (2) $A_i^{\theta_k} = \left[\begin{array}{c|c} rL_i^\theta & 0 \\ \hline 0 & 0 \end{array}\right]$ for $k < i < d$, (3) $A_k^{\theta_k} = \left[\begin{array}{c|c} L_k^\theta & C_k^\theta \\ \hline 0 & 0 \end{array}\right]$, and (4) $A_i^{\theta_k} = A_i^\theta$ for $i < k$, such that the loss along the path is at most $L(f_\theta) + \epsilon$. Note that θ_{d-1} satisfies all these assumptions, including in particular (2) as there are of course no A_i between A_{d-1} and A_d. Now let us extend this path to θ_{k-1}.

First, because the rightmost columns of A_i are zero for $k < i \leq d$, we can modify the bottom rows of A_i for $k \leq i < d$ without affecting the output of our network. In particular, we can set A_k to $\left[\begin{array}{c|c} L_k^\theta & C_k^\theta \\ \hline rL_k^\theta & 0 \end{array}\right]$, as well as A_i to $\left[\begin{array}{c|c} rL_i^\theta & 0 \\ \hline 0 & rL_i^\theta \end{array}\right]$ for $k < i < d$. From the fact that the loss is convex over A_d and that $L(f_{\theta_{k-1}}) < L(f_\theta) + \epsilon$, it then follows that we can set A_d to $\left[\begin{array}{c|c} 0 & rL_d^\theta \end{array}\right]$ via interpolation while keeping our loss below $L(f_\theta) + \epsilon$. In particular, note that because the off-diagonal blocks of A_i are zero for $k < i < d$, interpolating between the leftmost columns of A_d being non-zero and the rightmost columns of A_d being non-zero simply amounts to interpolating between the outputs of the two subnetworks comprised respectively of the first $\lfloor h_i/2 \rfloor$ and last $\lfloor h_i/2 \rfloor$ rows of A_i for $k \leq i < d$.

Once we have the leftmost columns of A_d set to zero and A_i in block-diagonal form for $k < i < d$, we can proceed to modify the top rows of A_k however we'd like without affecting the output of our network. Specifically, let us set A_k to $\left[\begin{array}{c|c} rL_k^\theta & 0 \\ \hline rL_k^\theta & 0 \end{array}\right]$. We can then reset A_d to $\left[\begin{array}{c|c} rL_d^\theta & 0 \end{array}\right]$ via interpolation—this time without affecting our loss since the weights of our two subnetworks are equivalent—and afterwards set D_k to zero and R_i to zero for $k \leq i < d$—again without affecting our loss since the rightmost columns of A_d are now zero, meaning that the bottom rows of A_i have no affect on our network's output.

Following these steps, we will have $A_i = \left[\begin{array}{c|c} rL_i^\theta & 0 \\ \hline 0 & 0 \end{array}\right]$ for $k \leq i < d$ and $A_d = \left[\begin{array}{c|c} rL_d^\theta & 0 \end{array}\right]$. And so we are now free to set the bottom rows of A_{k-1} to zero without affecting our loss, thus arriving at θ_{k-1}. $\square$

Lemma 4. *Let θ be a parameter such that at least $\lceil h_i/2 \rceil$ of the units in each hidden layer have been set to zero. Then we can achieve an arbitrary permutation of the non-zero hidden units of θ via a path consisting of just 5 line segments such that our loss is constant along this path.*

Proof. Let $\pi : [h_i] \mapsto [h_i]$ be some permutation over the units in layer i. Without loss of generality, suppose all non-zero units in layer i are indexed between 0 and $\lfloor h_i/2 \rfloor$, and define $\pi' : [\lfloor h_i/2 \rfloor] \mapsto [h_i] \setminus [\lfloor h_i/2 \rfloor]$ as any one-to-one mapping such that $\pi'(i) = \pi(i)$ if $\pi(i) \in [h_i] \setminus [\lfloor h_i/2 \rfloor]$. Note that when we refer to a unit j as "set to zero", we mean that both row j of A_i and column j of A_{i+1} have been set to zero.

To permute the units of layer i, we can first simultaneously copy the non-zero rows of A_i into a subset of the rows that have been set to zero. Specifically, for $j \in [\lfloor h_i/2 \rfloor]$ we can copy row j of A_i into row $\pi'(j)$ via interpolation and without affecting our loss, due to the fact that column $\pi'(j)$ in A_{i+1} is set to zero. We can then set column j of A_{i+1} to zero while copying its value to column $\pi'(j)$, again via interpolation and without affecting our loss since rows j and $\pi'(j)$ of A_i are now equivalent.

Following these first two steps, the first $\lfloor h_i/2 \rfloor$ columns of A_{i+1} will have been set to zero. Thus, for all $j \in [\lfloor h_i/2 \rfloor]$ such that $\pi(j) \in [h_i/2]$ we can copy row $\pi'(j)$ of A_i into row $\pi(j)$ without affecting our loss. We can then set column $\pi'(j)$ of A_{i+1} to zero while copying its value into column $\pi(j)$ via interpolation and without affecting our loss since rows $\pi'(j)$ and $\pi(j)$ of A_i are now equivalent. Setting row $\pi'(j)$ to zero—again for all $j \in [\lfloor h_i/2 \rfloor]$ such that $\pi(j) \in [h_i/2]$—completes the permutation for layer i.

Note that because we leave the output of layer i unchanged throughout the course of permuting the units of layer i, it follows that we can perform all swaps across all layers simultaneously. And so from the fact that permuting each layer can be done in 5 steps—each of which consists of a single line segment in parameter space—the main result immediately follows. □

Proof of Lemma 2. Without loss of generality, suppose for θ that the subset of $\lfloor h_i/2 \rfloor$ non-zero hidden units in each layer i are all indexed between 0 and $\lfloor h_i/2 \rfloor$. Note that when we refer to a unit as "set to zero", we mean that both the corresponding row of A_i and column of A_{i+1} have been set to zero. Adopting our notation in Lemma 1, we can construct a path from θ to θ' as follows.

First, from the fact that the second half of units in each hidden layer i have been set to zero in θ we have that $A_1^\theta = \left[\begin{array}{c} L_1^\theta \\ \hline 0 \end{array}\right]$, $A_i^\theta = \left[\begin{array}{c|c} L_i^\theta & 0 \\ \hline 0 & 0 \end{array}\right]$ for $1 < i < d$, and $A_d^\theta = \left[\begin{array}{c|c} L_d^\theta & 0 \end{array}\right]$. Similarly, half the rows of $A_1^{\theta'}$ are zero, half the rows and columns of $A_i^{\theta'}$ are zero for $1 < i < d$, and half the columns of $A_d^{\theta'}$ are zero. Note that the indices of the non-zero units in θ' may intersect with those of the non-zero units in θ. For $1 \leq i \leq d$, let B_i denote the submatrix of A_i corresponding to the non-zero rows and columns of $A_i^{\theta'}$.

Because A_i^θ are block-diagonal for $1 < i < d$ and the rightmost columns of A_d^θ are zero, starting from θ we can modify the bottom rows of A_i for $1 \leq i < d$ any way we'd like without affecting our loss—as done in our path construction for Lemma 1. In particular, let us set A_i to $\left[\begin{array}{c|c} L_i^\theta & 0 \\ \hline 0 & B_i^{\theta'} \end{array}\right]$ for $1 < i < d$ and A_1 to $\left[\begin{array}{c} L_1^\theta \\ \hline B_1^{\theta'} \end{array}\right]$. Then, from the fact that our loss function is convex over A_d it follows that we can set A_d to $\left[\begin{array}{c|c} 0 & B_d^{\theta'} \end{array}\right]$ via interpolation while keeping our loss below $\max\{L(f_\theta), L(f_{\theta'})\}$. Finally, from the fact that the leftmost columns of A_d are now zero and A_i are still block-diagonal for $1 < i < d$, it follows that we can set L_i to zero for $1 \leq i < d$ without affecting our loss—thus making A_i equal to $\left[\begin{array}{c|c} 0 & 0 \\ \hline 0 & B_i^{\theta'} \end{array}\right]$ for $1 < i < d$ and A_1 equal to $\left[\begin{array}{c} 0 \\ \hline B_1^{\theta'} \end{array}\right]$.

To complete our path from θ to θ' we now simply need to permute the units of each hidden layer so as to return the elements of $B_i^{\theta'}$ to their original positions in A_i for each i. From Lemma 4 it follows that we can accomplish this permutation via 5 line segments in parameter space without affecting our loss. Combined with the previous steps above, we have constructed path from θ to θ' consisting of a total of 8 line segments whose loss is bounded by $\max\{L(f_\theta), L(f_{\theta'})\}$. □

Proof of Theorem 1. First, from Lemma 1 we know we can construct paths from both θ^A to θ_1^A and θ^B to θ_1^B while keeping our loss below $L(f_{\theta^A}) + \epsilon$ and $L(f_{\theta^B}) + \epsilon$ respectively. From Lemma 2 we know that we